Type II

Technicians maintaining, servicing, repairing or disposing of medium, high, and very high-pressure appliances, except small appliances as defined in Type I and motor vehicle air-conditioning systems (MVAC), must be certified as a Type II Technician or a Universal Technician.

System Classifications

Air-conditioning appliances used to supply cooling in order to control heat and/or humidity in occupied facilities is defined by EPA as Comfort Cooling for the leak repair category. Residential, office, and commercial buildings are included in this category. Appliances (equipment) used to preserve products, as in the retail food and cold storage warehouse sectors are classified by EPA as Commercial Refrigeration.

When an appliance has a dual use, the applicable maximum leak rate is determined by the highest percentage of use. For example; If a company uses 55% of its systems cooling capacity to one of its manufacturing lines or refrigeration and 45% of its cooling capacity to air condition the office spaces, it is considered Industrial Process Refrigeration (IPR) for the leak rate category.

Determining System Charge

Use the charge stated on the equipment nameplate to determine normal charge for leak rate calculations when the system is a packaged system charged at the factory. The charge in field-piped split-systems can be calculated using the factory charge plus the charge added for piping and accessories. The leak rate of an appliance can be calculated when topping off or recharging a system due to a loss of charge.

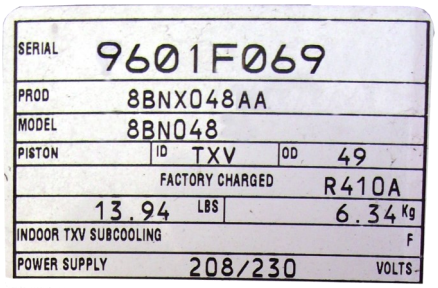

Figure 15. Data Plate.

The unit nameplate is the easiest way to check the type of refrigerant used in a packaged rooftop unit, Figure 15. In addition, the unit's nameplate will indicate the system's total charge. The best method to charge a system that has both an air-cooled condenser and evaporator with a nameplate charge of 80 pounds is a liquid charge by weight through the liquid-line service valve.

Type II Leak Repair Requirements

The EPA's leak repair requirements apply to all appliances using a regulated refrigerant. Once the threshold leak rate has been exceeded on any system with a charge size of 50 pounds or greater, the owner must repair the appliance enough to bring the leak rate below the threshold. If the owner cannot, or chooses not to, bring the leak rate below the threshold, the appliance must be retrofitted or retired.

Note: Leak repair requirements are the same for Type II & III.

As of January 1, 2019 the maximum leak rates are:

- 10% of the charge per year when used for comfort cooling.
- 20% of the charge per year when used for commercial refrigeration.
- 30% of the charge per year when used for industrial process refrigeration.

Leak Repair Time Frames

If an appliance with 50 or more pounds of a regulated refrigerant has exceeded the threshold leak rate, the equipment owner or operator has 30 days to repair the appliance so that it leaks below the threshold limit and conduct the initial verification test, unless EPA grants additional time. If an appliance with a charge of 200 pounds or more of a regulated refrigerant passes the initial verification leak test, a follow-up verification test must be conducted within 10 days. Not having a certified service technician available for service **cannot** be used as a reason to extend the appliance leak repair deadline.

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If after repairing a leak in an HCFC rack refrigeration system used in a supermarket, the system fails the initial leak verification test, the technician has a 30-day window to make additional repairs or develop a retrofit plan.

Assuming no extensions apply, a cold storage warehouse operating an appliance charged with R-22 or any other regulated substance leaking above the threshold rate, can continue to operate the appliance without repair for 12 months before it must be retrofitted or retired. If that system leak rate exceeds 125% of its full charge in a calendar year, then a report must be submitted to EPA describing the efforts to identify and repair system leaks. Further action must be taken by repairing, retiring, mothballing, or retrofitting the appliance.

Note: An appliance using an ozone-depleting refrigerant does not need to be removed or mothballed by storing in a warehouse at the facility to extend the leak repair deadline. The repair deadline can be extended by temporarily shutting down the appliance, recovering the refrigerant from an isolated section or the entire system to at least atmospheric pressure.

The owner or operator has 18 months to retrofit or retire a leaking appliance if the replacement appliance will use a refrigerant exempt from the venting prohibition.

Section 608 Leak Repair Regulations

The following chart summarizes the repair regulations as outlined above.

Note: Leak repair requirements are the same for Type II & III.

Section 608 Leak Repair Regulations		
The updated Section 608 regulations include new leak inspection and verification test requirements for owners/operators that affect technicians as of January 1, 2019:		
Leak inspections are required for appliances that have exceeded the maximum allowable leak rate, according to the schedule below. All visible and accessible components of an appliance must be inspected, using a method or methods that are appropriate for that appliance.		
Equipment	**Full Charge**	**Frequency of Required Leak Inspections After Repair**
Commercial Refrigeration and Industrial Process Refrigeration	**More than 500 pounds**	Once every three months until the owner/operator can demonstrate through leak rate calculations that the leak rate has not exceeded 20% (commercial refrigeration) or 30% (IPR) for four quarters in a row.
	50 to 500 pounds	Once per calendar year until the owner/operator can demonstrate through the leak rate calculations that the leak rate has not exceeded 20% (commercial refrigeration) or 30% (IPR) for one year.
Comfort Cooling	**50 or more pounds**	Once per calendar year until the owner/operator can demonstrate through the leak rate calculations that the leak rate has not exceeded 10% for one year.
Initial and follow-up verification tests of leak repairs are required for appliances that exceed the applicable leak rate. The verification tests must demonstrate that leaks were successfully repaired. *__An initial verification test__ must be performed before any additional refrigerant is added to the appliance. **__A follow-up verification__ test must be performed only after the appliance has returned to normal operating characteristics and conditions. There is no minimum timeframe. ***Technicians are required to** provide service invoices and records of leak inspections or verification tests to owners/operators.*		

Example: Based on the Section 608 Leak Repair Regulations chart, an industrial process refrigeration (IPR) system containing 1000 lbs. or more of R-22, that has exceeded the applicable leak rate threshold, must be inspected for leaks every 3 months until verification that leak rate has not been exceeded for one year after repair.

Record Keeping

The records of leak inspections, initial leak verification, and follow-up verification tests must be kept on file for three (3) years. The owner and/or operator of the equipment is responsible for keeping the records.

Leak Detection

After assembly and installation of a split-system, the unit should first be pressurized with an inert gas, such as nitrogen (may also be referred to as dry-nitrogen), and leak checked. When absolutely necessary, dry-nitrogen with a trace amount of the system's design refrigerant should be used for leak detection. For example, an R-407C system should be leak checked with pressurized dry-nitrogen and a trace amount of R-407C.

When first inspecting an operating hermetic system known to be leaking, you should look for traces of oil. Oil traces on the rotating shaft seal of a belt-driven compressor is an indication of a leak, especially if the compressor has not been used for several months. To pinpoint refrigerant leaks, testing with soap bubbles is best.

A refrigerant leak in a high-pressure capillary tube system should be suspected if the system operates with excessive superheat.

Recovery Techniques: Before You Recover

Although those who service Type II, stationary refrigeration appliances must own active recycling and recovery equipment, it is no longer required that the purchase of the equipment be reported to the EPA.

Before you recover refrigerant, determine the maximum amount of refrigerant that is contained in the system. This will be the system's total charge. If the total charge is greater than 15 pounds, system-dependent, or passive, recovery techniques **cannot** be used as EPA regulations prohibit their use when the appliance contains over 15 pounds of refrigerant.

Before using a recovery unit to remove a charge, the service valve positions and recovery oil level should be checked. When recovering a different refrigerant type than the last refrigerant recovered using that specific recovery equipment, remove any refrigerant remaining in the recovery unit and change the unit's filter. The recovered refrigerant may have contained impurities of acids, moisture, and oil.

The gauge hoses used to connect the recovery unit to the tank (and for all field service work) are required to have self-sealing connectors or hand valves to minimize refrigerant release when the hoses are connected or disconnected.

Before transferring refrigerant to an empty cylinder, the cylinder should be evacuated to reduce the pressure and remove non-condensables.

Before recovering refrigerant in a parallel piped compressor system, make certain that the equalization connections between compressors are closed. Recovery machines that utilize a water-cooled condenser are generally connected to the municipal water supply.

Recovery Techniques: Enhancing the Process

To reduce recovery time, connect one of the service hoses to the liquid line and recover refrigerant in the liquid phase or remove the refrigerant from the condenser outlet when the condenser is below the receiver. After the liquid refrigerant has been recovered, the remaining vapor is removed by the recovery system. Recovering refrigerant in the vapor phase, although slower than liquid recovery, will minimize oil loss from the appliance. Refrigerant that has been recovered from an air-conditioning system and held in a refillable cylinder may be charged back into the system after the repair. If, however, the appliance is being disposed of, it is important to know that all HVAC service companies must keep records for three (3) years for each appliance disposed of if the system's refrigerant charge is between 5 and 50 pounds.

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Cooling the recovery cylinder will lower its pressure and speed up the recovery process or removal of refrigerant charge from a system. In addition, heating the system will also speed up the recovery process as this will increase the pressure differential between the tank and the system. As the amount of refrigerant in the appliance drops, be careful to closely monitor the temperature of the recovery unit's compressor. This is because the hermetic compressor motors on recycling or recovery machines rely on the flow of refrigerant through the compressor for cooling and may overheat when drawing deep vacuums. In addition, the motor winding of a hermetic refrigeration compressor could be damaged if energized under a deep vacuum, so care should be taken to avoid energizing these compressors when refrigerant is being recovered.

Recovery Techniques: Required Evacuation Levels During Recovery

The required levels of evacuation during recovery are summarized in the following chart:

Required Levels of Evacuation During Recovery for Type II Appliances [Except for small appliances, MVACs, and MVAC-like appliances]	
Using recovery and/or recycling equipment manufactured or imported after November 15, 1993	
Very high-pressure appliance	0" Hg
High-pressure appliance, or isolated component of such appliance, with a full charge of less than 200 lbs. of refrigerant.	0" Hg
High-pressure appliance, or isolated component of such appliance, with a full charge of 200 lbs. or more of refrigerant.	10" Hg
Medium-pressure appliance, or isolated component of such appliance, with a full charge of less than 200 lbs. of refrigerant.	10" Hg
Medium-pressure appliance, or isolated component of such appliance, with a full charge of 200 lbs. or more of refrigerant.	15" Hg

Example 1: Before making a major repair to an R-410A (a high-pressure refrigerant) system that contains more than 200 pounds of refrigerant, the technician must evacuate the appliance (or component thereof) to a vacuum level of at least 10 inches of Hg vacuum.

Example 2: After reaching the required recovery vacuum level on an appliance, wait for a few minutes to see if the system pressure rises indicating moisture or a leak. For example, when changing out the compressor of a system containing 40 pounds of R-404A (a high-pressure refrigerant), evacuate the isolated section of the system to 0 psig and if system pressure does not rise after a few minutes, remove the compressor.

Example 3: If leaks in an appliance containing a regulated refrigerant make recovery to the prescribed level unattainable, the evacuation levels in the chart can be ignored, as the system refrigerant only needs to be recovered until the system's internal pressure reaches atmospheric pressure.

Note: *Depending on the question, the EPA exam may use the word* ***EVACUATION*** *in place of* ***Dehydration*** *or in other questions in place of* ***Recovery***.

Refrigeration System Liquid-Line Accessories

The refrigerant receiver will typically be found in the liquid line, at the outlet of the condenser, when the system uses a thermostatic expansion valve as its metering device. When the system is operating, the state of refrigerant leaving the receiver is a high-pressure liquid. Prior to opening a system with a liquid-line receiver for repair, the system's refrigerant should be pumped down and isolated in the receiver whenever possible to eliminate the need to recover the refrigerant from the system.

A moisture indicating sight glass is used for checking refrigerant for moisture. The sight glass is located in the liquid line, immediately downstream, after, the liquid-line filter drier. Anytime a system is opened for service, the liquid-line filter drier should be replaced. The filter drier is used to remove moisture from the refrigerant in a system.

Note: Isopropyl alcohol should be used to remove ice from sight glasses or viewing glasses.

Refrigeration System Suction Line and Compressor Accessories

In the suction line, at the outlet of the evaporator, suction line accumulators can be found. They are located between the evaporator and the compressor. Accumulators prevent any liquid refrigerant that may flow from the evaporator from flowing into the compressor. The normal position of the suction service/shutoff valve is back-seated. In the back-seated position, the gauge port is closed off so the manifold hose can be removed from the valve without releasing refrigerant. On the high side of the compressor, the discharge valve can be found. As with the suction service valve, the normal operating position is back-seated. Under no circumstances should the discharge service valve be closed on an operating reciprocating compressor. Doing so can result in compressor damage as well as severe personal injury. To help prevent compressor damage, refrigerant systems must be protected by a high-pressure relief device which may be in the compressor or external relief valve. When using multiple pressure relief valves they can only be installed in parallel.

Crankcase heaters are used on refrigeration and air-conditioning compressors to reduce the amount of refrigerant in the oil and to reduce oil foaming which usually occurs in the compressor crankcase. Should the compressor burn-out, an oil sample should be taken to test for the presence of acid in the system. Compressor replacement, as well as the replacement of an evaporator coil, condenser coil, or other system heat exchanger, is considered to be a “major” repair as defined by EPA regulations.

System Evacuation

A deep vacuum is measured in microns. A vacuum level of 500 microns should be achieved to ensure all moisture and non-condensables have been removed from a refrigeration system. When evacuating a system with large amounts of moisture it may be necessary to break the vacuum by increasing the pressure with nitrogen to counteract freezing of the moisture. Water in the system can freeze when using too large of a vacuum pump to evacuate a system. If non-condensables, such as air or dry-nitrogen, remain in a refrigeration system, the discharge pressure of the system will be high.

Notes: Nitrogen tanks are under high pressures, so the nitrogen cylinders require a pressure regulator to reduce the pressure to a safe level.

On a refrigeration system, never install multiple pressure relief valves in series.

System Charging and the Pressure-Temperature Chart

In an R-410A chilled-water system, vapor is charged from a vacuum level to a pressure of approximately 110 psig, which corresponds to a temperature of 36°F, to prevent water freezing. Similarly, the saturation pressure of an R-410A machine at (80°F) room temperature, while the machine is idle, is 238 psig when referring to a PT chart. An R-134a chiller should be initially charged with vapor to a pressure of approximately 30 psig, which corresponds to a temperature of 35°F. The evaporation temperature of R-134a at 0 psig is -15°F. For R-123, a low-pressure refrigerant that is used in low-pressure appliances, the saturation pressure at 35°F is 19.5”Hg vacuum (Refer to the Core section for information on how to use the PT chart). ASHRAE Standard 15 requires the use of sensors to detect the presence of refrigerant with few exceptions. ASHRAE Standard 15, for example, does not require an equipment room refrigerant detector for R-717 provided the mechanical ventilation system in the machinery room is run continuously.

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